

Nonlinear Absorbing-Loop Mirror Mode-Locked all-Polarization-Maintaining Yb-Doped Fiber Laser

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Abstract—In this paper, a nonlinear absorbing-loop mirror (NAbLM) is constructed in a polarization-maintaining (PM) version and applied in the all-normal dispersion (ANDi) regime, for the first time to our best knowledge. To investigate the properties of NAbLM in contrast with typical nonlinear optical loop mirror (NOLM), an asymmetric optical coupler is used in the NAbLM. In experiment, both the NAbLM and NOLM can be used to mode-lock a Yb-doped fiber (YDF) laser and enable a type of noise-like pulse (NLP) generation. Noticeably, however, it is further found that, in comparison to NOLM, NAbLM evidently enables both temporal and radio frequency (RF) noise suppressions, as well as elimination of RF variations. Further calculations show that the saturable property of the incorporated absorber plays a vital role in differing the NAbLM from typical NOLM. Our results demonstrate some new features of NAbLM, especially in respect to temporal pulse shaping, indicating that NAbLM can be a useful type of fiber loop mirror in nonlinear fiber optics.

Index Terms—Fiber laser, nonlinear loop mirror, polarization maintaining.

I. INTRODUCTION

NONLINEAR optical loop mirror (NOLM) and nonlinear amplifying-loop mirror (NALM), well known as two basic Sagnac fiber loop mirrors, have been widely used in nonlinear fiber optics. They are particularly useful as artificial saturable absorbers (SAs) in mode-locked fiber lasers. Besides applications in ultrashort pulse generation [1], [2], these devices have also been used to realize some long-duration pulse generation with peak-power-clamping (PPC) effect. Those include rectangular-shaped pulse [3], [4], h-shaped pulse [5], chair-like pulse [6], etc. Among the reported rectangular-shaped pulses, most were dissipative soliton resonance (DSR) pulses [7]–[13], but some belonged to a type of noise-like pulse (NLP) [3], [14]–[16]. Autocorrelation trace (ACT) of an NLP typically exhibits an ultrashort peak anchored on a relatively broad pedestal [17]–[19].

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Despite the NOLM and NALM, several other variants have also been investigated for potential applications in optical signal processing or nonlinear fiber optics. Those include simultaneous incorporation of an optical amplifier and an optical attenuator in a loop mirror [20], NOLM with-imbalanced lumped loss [21], [22], NOLM with a directional phase shifter [23], as well as the recently reported NAbLM [24].

In this paper, we first construct NAbLM in an all polarization-maintaining (PM) version in a Yb-doped fiber (YDF) laser, and also for the first time apply it in an all-normal dispersion (ANDi) regime. Different to that in Ref. [24] where an evenly-splitting fiber optical coupler (FOC) was used, here we use a highly asymmetric PM-FOC to construct the NAbLM. This is mainly in purpose to investigate what differences or merits might be with the NAbLM in comparison to typical NOLM. Experimentally, here the PM-YDF laser can produce a type of NLP by using either the NOLM or the NAbLM. It is found that using the NAbLM evidently produces NLP in a more ordered manner. We further take some further theoretical analysis and simulations on the differences between the NAbLM and NOLM.

II. EXPERIMENTAL DETAILS

The Experimental layout of the whole PM YDF laser is schematically depicted in Fig. 1, which includes a unidirectional loop (UL) and a bidirectional NAbLM. The UL and NAbLM are connected by using a 2×2 PM fiber optical coupler (PM-FOC) with splitting ratio of 5:95. In the UL, two single-mode laser diodes (LDs) with center wavelength at 976 nm are used as the forward and backward pump sources, respectively. The pump lights are launched into a piece of ~ 1.45 m PM-YDF via two PM wavelength division multiplexers (PM-WDMs). The PM-YDF has a specified core absorption at 975 nm, core diameter, and core numerical aperture of 250 dB/m, 6 μ m, and 0.11, respectively. A PM circulator (PM-CIR) is used to enable unidirectional propagation of the signal laser in the UL. The PM-CIR can simultaneously act as the output coupler by ejecting a portion of the returned light from the PM-FOC. Here, using the PM-CIR to replace typical “isolator+FOC” combination offers at least two benefits. At first, it makes the system more compact by using one component to replace two. Secondly, all the returned light from the NAbLM will either be ejected as output or further amplified in the gain fiber. This is quite different from typical “isolator+FOC” combination, where a portion of the returned light provides the output but the left portion is either dissipated or directed out at the isolator.

In the NAbLM, a piece of the same type of PM-YDF with length of ~ 0.36 m is used as the absorber. It should be noted that this piece of PM-YDF is spliced to the loop closer to the 5% port of the PM-FOC, regarding to the incident wave from the UL to the NAbLM. This results in an adding-up between the asymmetries induced by the PM-FOC itself and the unpumped PM-YDF. The two guided counter-propagating

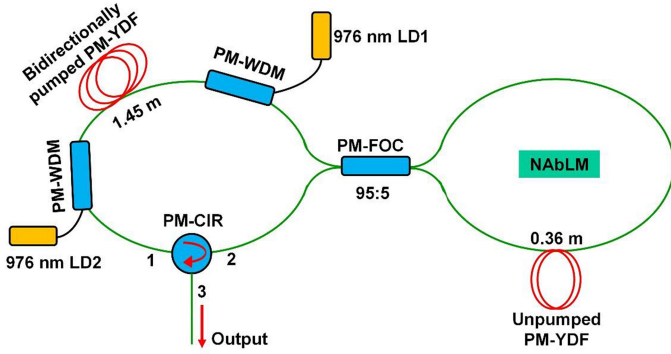


Fig. 1. Experimental layout of NAbLM-mode-locked PM YDF laser.

waves from the PM-FOC will both be absorbed by the incorporated PM-YDF, but not simultaneously: One is absorbed just after entrance to the loop while the other counterpropagating one experiences absorption just before exiting the loop. Thus, almost throughout the NAbLM the two waves differ by a large amount in intensity. The difference is even greater if the high asymmetry of the used PM-FOC itself is further taken into account. Consequently, a large amount of differential nonlinear phase shift can be induced.

Apart from above mentioned fibers and components, all other connecting or pigtail fiber pieces are the same type of single mode PM fiber (PMF, PM980-XP, Nufern, Inc.). As measurements, the lengths of the NAbLM and the overall cavity are ~ 2.4 m and ~ 8.3 m, respectively. The net cavity dispersion is ~ 0.19 ps².

It should be noted that here we used a highly asymmetric FOC in the NAbLM. Thus, even without the absorber, the YDF laser is still able to attain mode-locking if suitable parameter manipulations are implemented. In that case, the NAbLM simply reduces to a typical NOLM. The reason why we adopt such an asymmetric layout is based on the consideration that here we mainly intend to investigate the property differences with the NAbLM in comparison to typical NOLM. Another point is that our YDF laser operates in the ANDi regime, different to the NAbLM-incorporated Ho-doped fiber lasers in Ref. [20] that operated in either anomalous dispersion regime or net-normal dispersion regime.

III. RESULTS

Fig. 2 comparatively shows the measured spectral and temporal characteristics when the NAbLM and its corresponding NOLM are used, respectively. For the NOLM, the incident pump powers from LD1 and LD2 are ~ 486 and ~ 287.7 mW, respectively, yielding an average output power of ~ 44.1 mW. For the NAbLM, the mode-locking threshold is slightly higher, mainly due to the additional absorption by the unpumped PM-YDF. The used powers from LD1 and LD2 are then ~ 486 and ~ 298.3 mW, respectively. Meanwhile, the average output power was ~ 41.6 mW, slightly lower than that with the NOLM. It should be noted that a piece of PM980-XP with approximately the same length of the PM-YDF have been incorporated additionally in the NOLM-based experiment, intending to compensate the length difference between the two laser cavities.

The spectra through this paper are all measured by using the same optical spectrum analyzer (AQ6370D, Yokogawa Test & Measurement Co., Japan) with the same setting resolution of 0.02 nm. Fig. 2(a) comparatively plots the output spectra when the NAbLM and NOLM are used. As seen, in contrast with the NOLM, some short-wave components disappear when the NAbLM is used. This is mainly

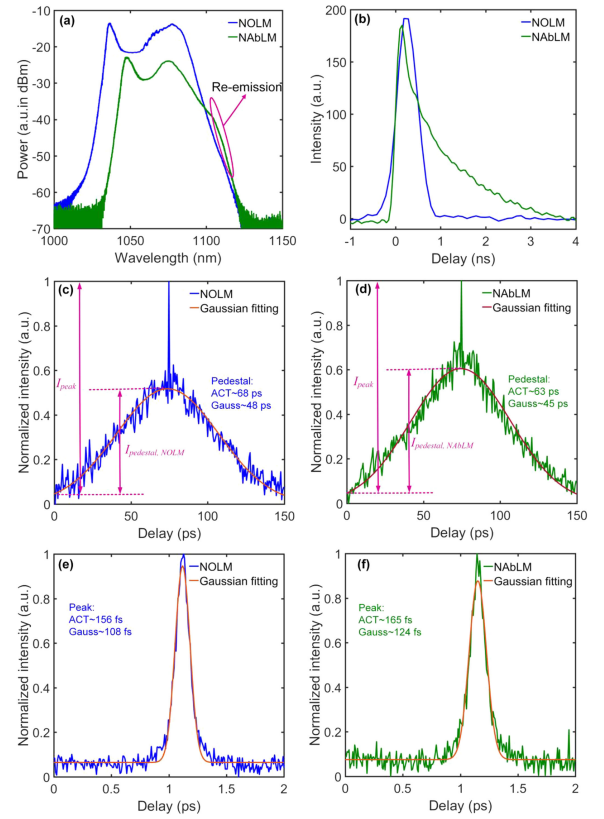


Fig. 2. Mode-locking characteristics with the NOLM and NAbLM, comparatively: (a) spectral profiles; (b) single pulse envelopes; overall ACTs with Gauss fits when using (c) NOLM and (d) NAbLM, respectively; ACTs around the coherence peaks with Gauss fits when using (e) and (f), respectively.

due to the absorption by the YDF incorporated in the NAbLM. But meanwhile some new spectral components appear beyond the original long-wave side, which should result from re-emission in the YDF after absorption. Re-emission effect is significant and able to dominate the spectral characteristics even only with small-signal absorption [25]. Besides these differences, one common feature is that both the spectra span broadly, exhibiting 10-dB bandwidths over 50 nm.

Fig. 2(b) plots the temporal characteristics measured by using a fiber-coupled InGaAs PIN photodiode (PD) (ET3500F, Electro-Optics Technology, Inc.) connected to a digital oscilloscope (WaveRunner 8404M, Teledyne Lecroy, Inc.). Limited by the response time of the PD, the pulse characteristics could not be identified accurately. Even that, as seen in Fig. 2(b), when the NAbLM was used, the pulse leading edge rose faster than that when the NOLM was used. This should attribute to the additional absorption by the YDF in the NAbLM. In detail, the unpumped PM-YDF itself was able to function as an artificial SA, which tended to eliminate some weak light. Thus, the pulse would leave a steep leading edge, especially after numerous roundtrips of circulation.

Figs 2(c) and 2(d) plot autocorrelation traces (ACTs) with the NOLM and NAbLM, respectively, measured by using a commercial autocorrelator (PulseCheck 600, APE, GmbH, Germany). As seen, each ACT contains a high-rising ultrashort peak standing out from a spike-embedded, much longer pedestal. These are just characteristics of typical NLPs, relating to their intrinsically partial and low coherence [7], [26]. The ratio R between the peak intensity I_{peak} and pedestal intensity $I_{pedestal}$ can be used to roughly account for the ratio between the noise-like part intensity and the average pulse

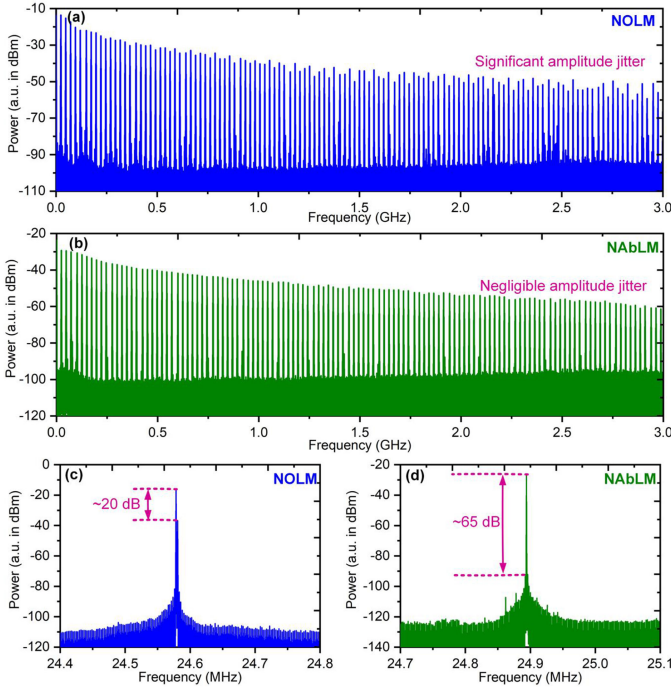


Fig. 3. RF trains spanning from 0 to 3 GHz with (a) NOLM and (b) NAbLM; RF trace around the PRF with (c) NOLM and (d) NAbLM, respectively.

intensity.

$$R = I_{\text{peak}} / I_{\text{pedestal}} \begin{cases} = 1, & \text{noise-free pulse,} \\ > 1, & \text{NLP.} \end{cases} \quad (1)$$

If $R = 1$, the pulse is a noise-free pulse. In case that $R > 1$, the pulse is a partially-coherent and noise-distorted pulse, i.e., a typical NLP. For a very large value of R , i.e., $R \rightarrow \infty$, the pulse approaches a stochastically noise-filled pocket.

As for our laser operation with the NOLM [Fig. 2(c)], $R = \sim 2.04$, the generated pulse is then a typical NLP. For the NAbLM, notably, R reduces to ~ 1.70 [Fig. 2(d)]. This indicates that the noise-like part decreases and the temporal structure of the NLP tends to become more ordered.

Figs 2(e) and 2(f) zoom in peak part of each NLP. The peak part results from the weak coherence among the ultrashort internal pulses in each NLP. As seen, each peak is short down to roughly 100 fs, which results from the short coherence length among the internal pulses in each NLP. However, it can be noted that the coherence length with the NAbLM (~ 124 fs) is slightly longer than that with the NOLM (~ 108 fs). This further lays an evidence that the NAbLM enables the NLP to be temporally more ordered in contrast with its counterpart of NOLM.

Fig. 3 plots radio frequency (RF) characteristics with both the NOLM and NAbLM. These are all measured by using aforementioned PD in combination with an RF spectrum analyzer (N9020B, Keysight, Inc.). For each trace with 0-3 GHz span, the resolution bandwidth (RBW) is set at 5.1 kHz, whereas for each trace around the fundamental pulse repetition frequency (PRF) the RBW is set at 1 Hz. As noticed in Fig 3(a), the 0-3 GHz RF trace with the NOLM shows significant variations, especially at high order RF harmonics. Strikingly, however, the RF trace drops smoothly without noticeable variations when the NAbLM is used, as seen in Fig. 3(b). Although there is no reason to prefer pulses with a monotonically decreasing Fourier transform for the internally complex NLP, using the NAbLM can indeed make the RF variations much less

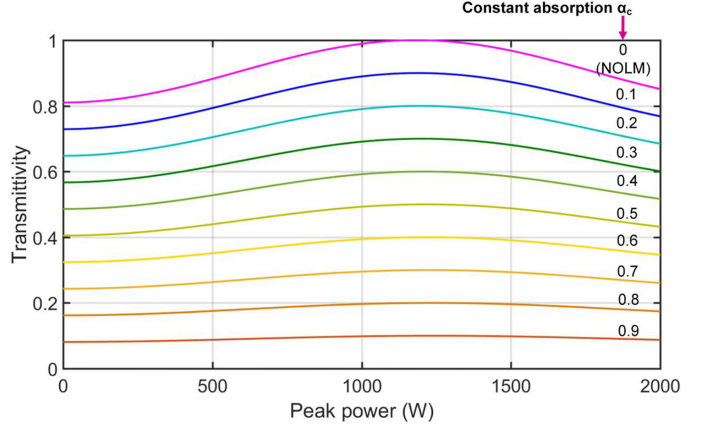


Fig. 4. Peak-power-dependent transmittivity of NAbLM with different values of constant absorption.

significant in contrast to using the corresponding NOLM. One probable factor might be that the NLP is now much more ordered due to the noise-suppressing effects with the NAbLM.

Figs 3(c) and 3(d) plot the RF traces around the fundamental PRF with the NOLM and NAbLM, respectively. As a common feature, there are some side RF peaks appearing at the vicinities of the main RF peaks. For the NOLM, the main-to-side contrast of peak powers, i.e., signal-to-noise ratio (SNR), can only reach ~ 20 dB. Comparatively, the NAbLM can enhance the SNR to ~ 65 dB, centering at ~ 24.894 MHz. Thus, despite elimination of the RF amplitude fluctuations as previously observed, the NAbLM is also able to partially suppress RF noises around the fundamental PRF.

All above observed spectral, temporal, and RF characteristics with the NAbLM should all attribute to the introduced additional absorption by the unpumped PM-YDF. Below, at first we only give some brief analysis by assuming that the absorption coefficient α of the PM-YDF is a constant value α_c .

Here, we define α as the ratio between the absorbed power through the absorber and the total incident power,

$$\alpha = \frac{P_{\text{in}} - P_{\text{out}}}{P_{\text{in}}} \quad (2)$$

Thus, now the transmittivity T_s of the NAbLM can be written as

$$T_s = 1 - \alpha - 2(1 - \alpha)\rho(1 - \rho) \times \{1 + \cos[(1 - 2\rho + \alpha\rho)\bar{\gamma}P_{\text{peak}}L]\}, \quad (3)$$

where ρ , $\bar{\gamma}$, P_{peak} , and L are splitting ratio of the FOC, averaged fiber nonlinear coefficient in the NAbLM, pulse peak power, and loop length, respectively. Substituting $\alpha = \alpha_c$, $\rho = 0.05$, $\bar{\gamma} \approx 1.23 \text{ W}^{-1}\text{km}^{-1}$ [27], and $L = 2.4$ m into Eq. (3), we obtain

$$T_s = (1 - \alpha_c) \{0.905 - 0.095 \cos[(2.6568 \times 10^{-3} - 1.476\alpha_c \times 10^4)P_{\text{peak}}]\}, \quad (4)$$

Fig. 4 plots peak-power-dependent transmission characteristics of the NAbLM for some different values of constant absorption factor α_c . It can be seen that both the transmittivity and the modulation depth become less as the increase of the amount of incorporated absorption. This indicates that the NAbLM becomes even worse in comparison to the corresponding NOLM. However, this is inconsistent with our aforementioned experimental results considering that significant differences by using the NAbLM have been achieved evidently.

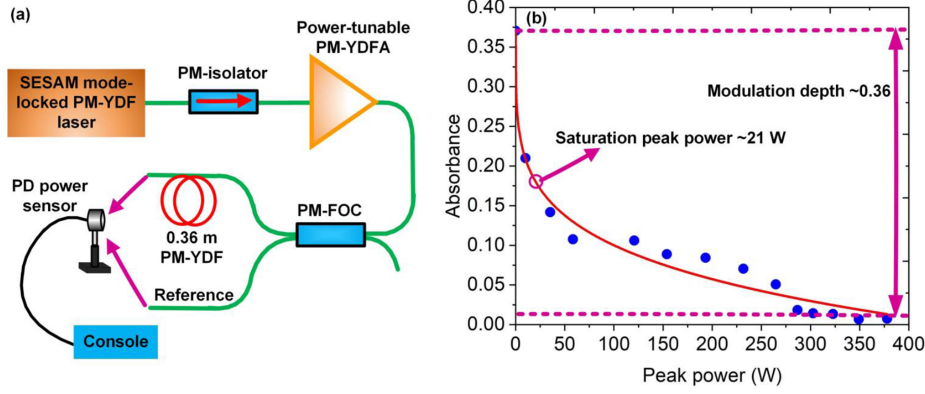


Fig. 5 (a) Experimental setup for peak-power-dependent absorption measurement of the unpumped PM-YDF; (b) measured absorbance against the incident pulse peak power with nonlinear fitting.

To further understand such inconsistency, we further consider the power-dependence of the absorber incorporated in the NAbLM, i.e., the unpumped PM-YDF itself is considered as an SA. For that, α should be seen as a function of P_{peak} , no more a constant. Considering the general case with saturable absorption and neglecting the nonsaturable loss for simplicity, $\alpha(P_{peak})$ can be written as

$$\alpha(P_{peak}) = \alpha_0 \left(1 + \frac{P_{peak}}{P_{sat}} \right)^{-1}, \quad (5)$$

where $\alpha_0 \in (0, 1)$ is the small-signal absorption, equivalent to modulation depth, and P_{sat} is the saturation power. By convention, P_{sat} is defined as the pulse peak power when the absorption reduces to half its small-signal value [28].

To identify the α_0 and P_{sat} of the used PM-YDF, we further take a power-dependent absorbance measurement performed with a pulsed fiber laser source. Fig. 5(a) shows the measurement setup. A house-made PM-YDF laser mode-locked by using a semiconductor saturable absorbing mirror (SESAM) is implemented as the seeding source. It can deliver pulses at a center wavelength, pulse duration, PRF of ~ 1063.75 nm, ~ 10.6 ps, and ~ 19.67 MHz, respectively. The output average power is ~ 10 mW, which can be further amplified in a PM-YDF amplifier (PM-YDFA) up to >80 mW. Then the pulses are split into two portions through a PM-FOC. One portion first goes through the previously used PM-YDF and then targets at a power meter containing a photodiode (PD) power sensor a connected digital console, whereas the other portion directly targets at the same power meter as a reference. The measured result is shown in Fig. 5(b). Based on nonlinear fitting, the modulation depth $\alpha_0 = 0.36$ and the saturation peak power $P_{sat} = 21$ W. Substituting these values into Eq. (5),

$$\alpha(P_{peak}) = 0.36 \left(1 + \frac{P_{peak}}{21 \text{ W}} \right)^{-1} \quad (6)$$

Using $\alpha(P_{peak})$ in Eq. (6) to replace α_c in Eq. (4), we obtain

$$T_s = \left[1 - 0.36 \left(1 + \frac{P_{peak}}{21 \text{ W}} \right)^{-1} \right] \cdot \left\{ 0.905 - 0.095 \cos \left[2.6568 \times 10^{-3} P_{peak} - 1.476 \times 10^{-4} \times 0.36 \times \left(\frac{1}{P_{peak}} + \frac{1}{21 \text{ W}} \right)^{-1} \right] \right\}. \quad (7)$$

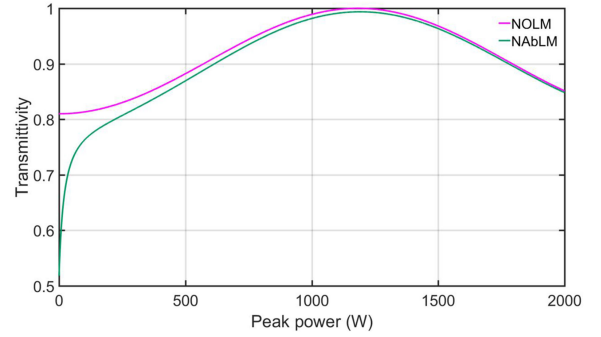


Fig. 6. Peak-power-dependent transmittivity of SA-incorporated NAbLM.

The calculated results based on Eq. (7) are shown in Fig. 6. Now, comparatively the transmittivity evolution of the NAbLM is considerably different from that of the NOLM. As seen, one is that the transmittivity increases much more rapidly as the incident pulse peak power approaching saturation. The second is that the modulation depth of the NAbLM is considerably greater than that of the NOLM. Although comparatively there is some decrease with the transmittivity due to the incorporation of the SA in the loop, the amount of decrease is substantially weaker in contrast with the amount of the enhanced modulation depth especially when the transmittivity approaches saturation. It should be noted that here we only need to pay attention to the rising part of the curves plotted in Fig. 6, considering that now the NAbLM is only used as an SA, not a limiter. These contrasts shown in Fig. 6 indicate that the NAbLM should exhibit stronger pulse shaping capability than the pure NOLM. This should be the main mechanism accounting for previously observed suppression of temporal and RF noise, as well as the elimination of RF variations.

Another notable benefit with NAbLM is that the incorporated fiber absorber itself is made of real material. A real-material-based SA intrinsically has better self-starting capability, considering that its absorption is always intensity-dependent even starting from zero. Whereas for typical NOLM or NALM, intensity dependence of the absorption cannot start from zero. This is due to that these loops are strictly reciprocal for low-peak-power pulses [29]. Thus, for incident weak pulses, these loops function as reflective mirrors that can reflect the pulses back to the same port rather than transmit them through the other port. In such case, neither the NOLM nor NALM can offer effective intensity dependence of transmission. That is why these devices typically require an additional optical component that enables some amount of

nonreciprocal phase bias, in order to achieve intensity dependence even from zero, and then enable better self-starting.

Besides the NLPs, from a general point of view the NAbLM-based fiber lasers should be able to produce some other typical ultrashort pulses. These might include soliton, dissipative soliton, dispersion-managed soliton, self-similar pulse, etc., depending on dispersion and nonlinearity circumstances. As for our current laser system in the ANDi regime, the conventionally realized pulse should be dissipative soliton, but in fact it can only generate NLPs in our experiment. One possible factor should be the lack of spectral-filtering mechanism to stabilize the highly chirped pulse in the ANDi regime [30]–[34]. A proper spectral filter facilitates the intracavity dynamics evolving into an equilibrium and stabilized state, which is typically essential for ANDi or even dispersion-managed mode-locked fiber lasers [35]. Such filter-incorporated investigation on NAbLM-mode-locked fiber lasers might be one of our future considerations.

IV. CONCLUSION

In conclusion, we have experimentally achieved passive mode-locking of an all-PM, ANDi, YDF laser by using an NAbLM as the artificial SA. The NAbLM can enable significant suppression in both temporal and RF noises, as well as elimination in amplitude fluctuations of RF. We further theoretically calculated the significantly different transmission characteristics of the NAbLM from its counterpart of NOLM. Our results demonstrate that NAbLM could be a useful device in ultrafast and nonlinear fiber optics, exhibiting some experimentally and theoretically verified benefits in comparison to typical NOLM and thus deserving to be further investigated.

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